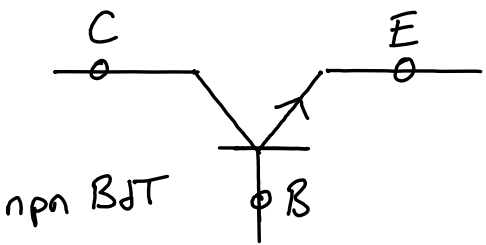
Bipolar Junction Transistor (BJT)



Collector collecting carriers

Base

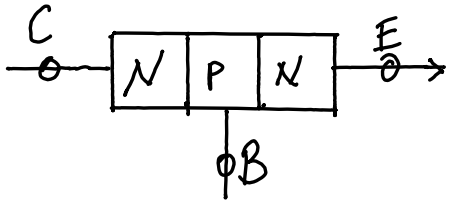
Emitter: Emitting carriers

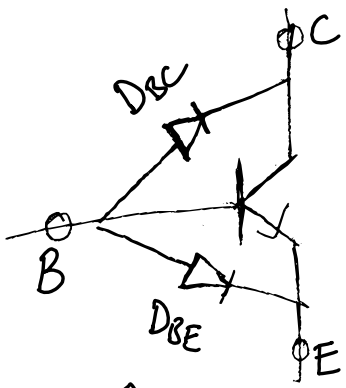


npn BJT

Applications:

1. Amplifier
2. Switch



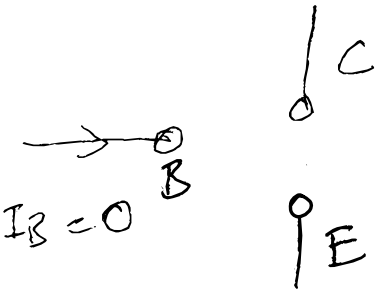


⇑

Analogy

D_{BE}	D_{BC}	Transistor Mode
OFF	OFF	OFF
ON	OFF	Forward Active (FA)
OFF	ON	Reverse Active (RA)
ON	ON	SAT (Saturation)

OFF



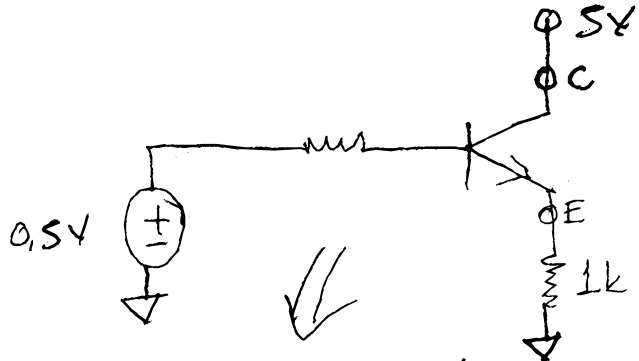
conditions:

$V_{BE} < V_{BE}$ (ON) Turn on Voltage for BE

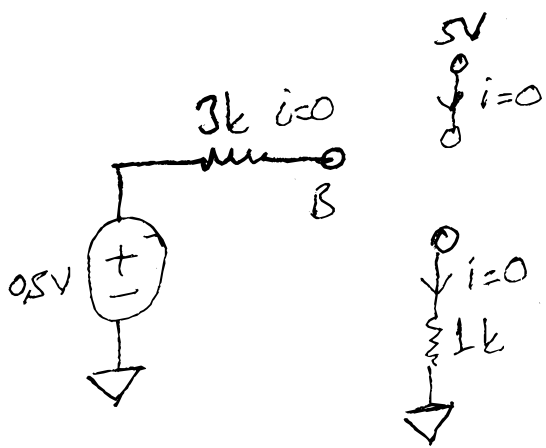
$V_{BC} < V_{BC}$ (ON)

Example:

$V_{BE}(ON) = V_{BC}(ON) = 0,7\,V$



⟷ Assume OFF



$V_B = 0,5 - 0 = 0,5V$

$V_E = 0V$

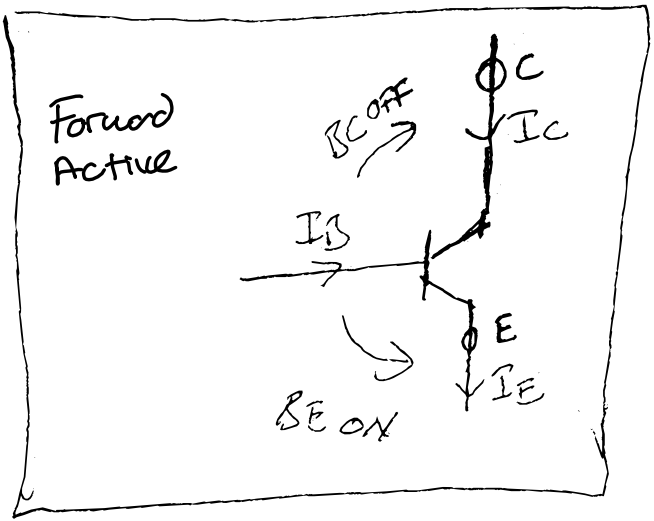
$V_C = 5V$

$V_{BE} = 0,5V < V_{BE}(ON)$

$V_{BC} = 0,5 - 5 = -4,5V$

$< V_{BC}(ON)$

✓ Assumption is correct



$$I_C = \beta_{FA} * I_B$$

$$I_E = (\beta_{FA} + 1) I_B$$

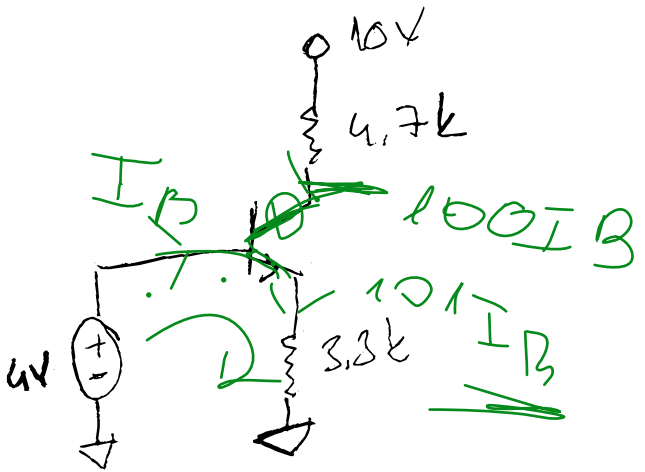
Conditions:

$I_B > 0$

$V_{BC} < V_{BC(ON)}$

V_{CE} is positive

Example



$\beta_{FA} = 100$

$V_{BE(on)} = (on) = V_{BC(on)} = 0,7V$

$10 - 4.7(100 I_B) \overset{?}{=} V_C$ $\quad I_B > 0V$

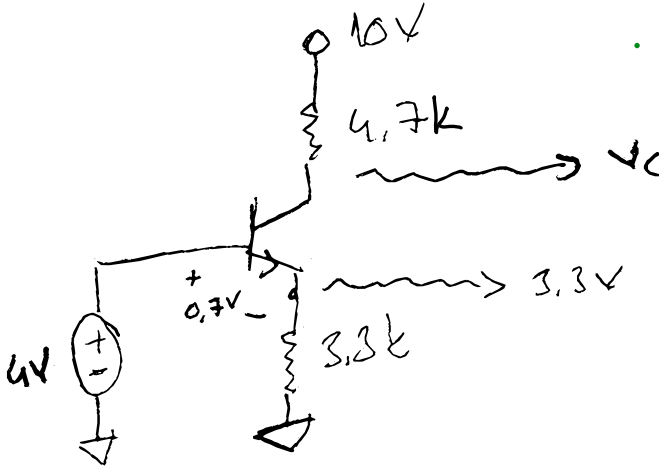
$-0.7 - 3.3(10 I_B) = 0$

$V_{BC} = V_B - V_C < 0.7V$

$4 - 5.3$

Example:

Forward Active Assumption



$V_C = 10 - I_C \times 4.7k$

$V_C = 5,3V$

$I_E = 1mA$

$I_B = \frac{1mA}{(\beta_{FA}+1)} = \frac{1mA}{101}$

$I_C = \beta_{FA} \times I_B = 0,99mA$

check FA conditions

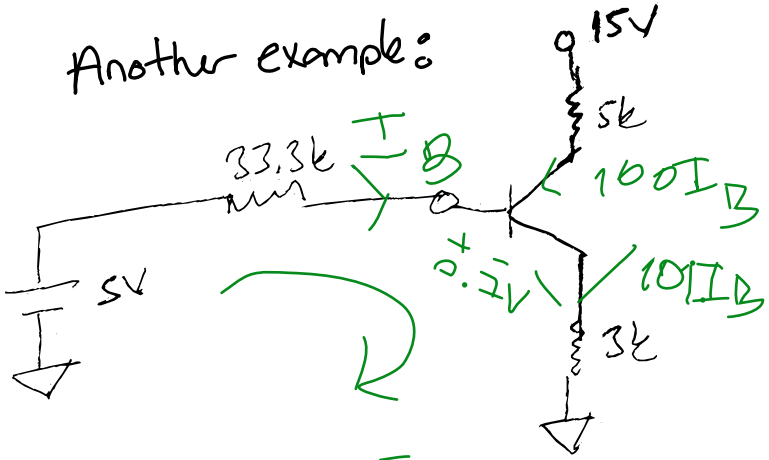
(1) $I_B > 0$

(2) $V_{BC} < V_{BC(on)}$

$4 - 5,3 = -1,3 < 0,7$

Assumption is correct ✓

Another example:



$\beta_{FA} = 100$

Turn-on = 0.7V

$5 - 33.3kI_B - 0.7 - 101 \times 3k = 0$

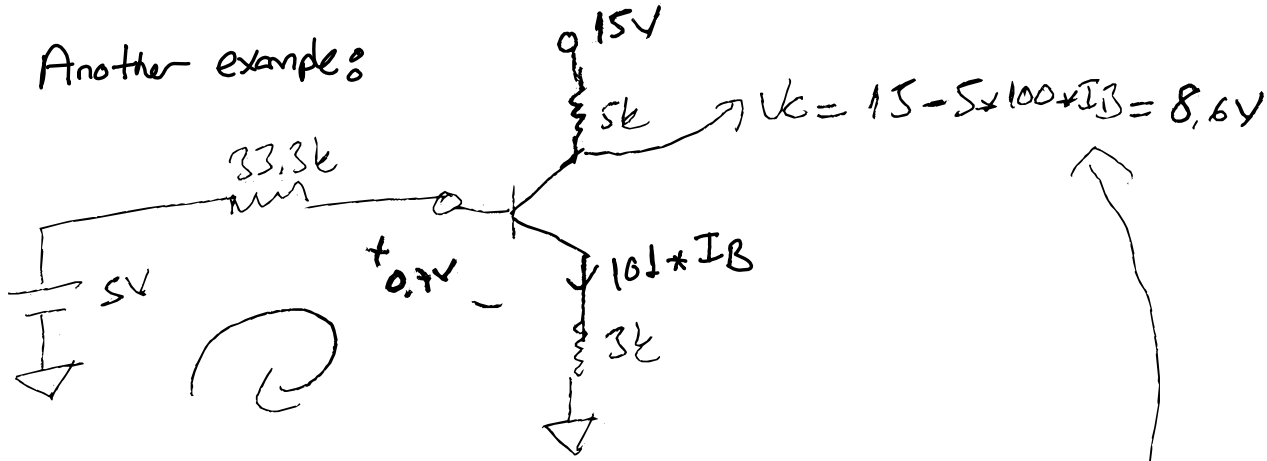
$I_B = 0.0128mA$ ✓

$V_C = 15 - 5k \times (100I_b) = 8.6V$

$V_B = 5 - 33.3kI_B = 4.3V$

$V_{BC} = 4.3 - 8.6 < 0.7$ ✓

Another example:



KVL: $5V - 33,3k * I_B - 0,7V - 3k * 101 * I_B = 0$

$$I_B = \frac{336,3}{43} mA$$

$$= 0,0128 mA$$

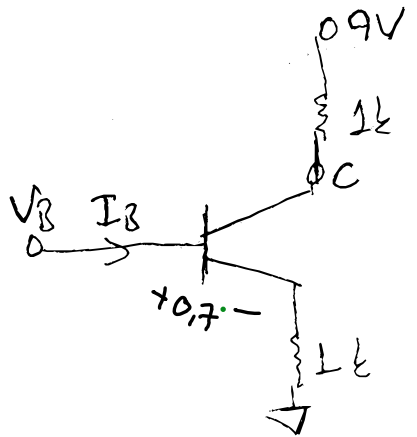
check FA:

$I_B > 0$ ✓

$V_B = 5 - (33,3) * I_B = 4.03 \Rightarrow V_{BC} = -4,57V$

$< 0,7$ ✓

Example



Transistor has a high β_{FA}.

Find the max value for V_B such that transistor is in FA.

$$I_E = \frac{V_B - 0,7V}{1k} = (V_B - 0,7)\,mA$$

$$I_B = \frac{I_E}{\beta_{FA}+1} = \frac{V_B - 0,7}{\beta_{FA}+1}\,mA$$

for FA $I_B > 0 \Rightarrow V_B > 0,7V$

$$V_{BC} = V_B - (9 - I_C \cdot 1k)$$

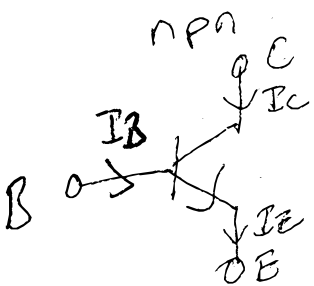
$$= V_B - 9 + \beta_{FA} \cdot \frac{I_E}{\beta_{FA}+1} \sim V_B - 0,7 \text{ as } \beta_{FA} \text{ is high}$$

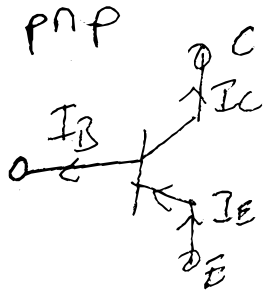
$$= V_B - 9 + V_B - 0,7 = 2V_B - 9,7$$

To find max. val. in FA $\Rightarrow V_{BC} < V_{BC(ON)} \Rightarrow 2V_B - 9,7 < 0,7 \Rightarrow V_B < 5.2$

SUMMARY

$I_C \sim I_E$ $\quad \alpha = \frac{I_C}{I_E} \sim [0{,}98,\ 0{,}99]$



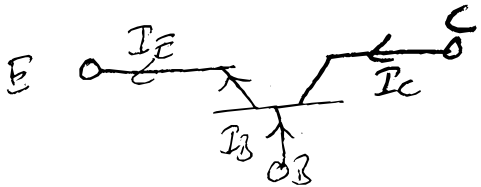


$I_E = I_C + I_B$ $\quad I_C = \beta * I_B$ $\quad I_E = (\beta + 1) * I_B$

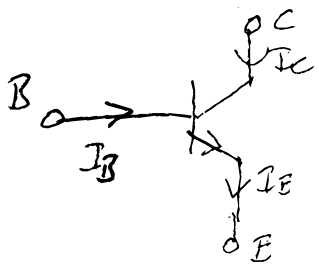
(for active region)

Configurations

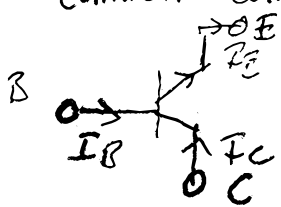
Common-Base npn



Common-Emitter

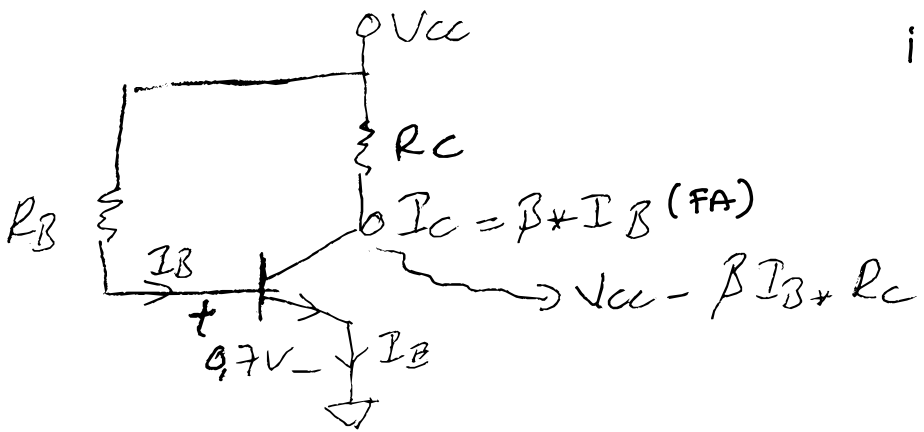


Common-Collector



from FA to SAT

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$



if we increase I_B further the transistor tries to increase I_C but it has a limit

$$I_C < \beta I_B$$

R_B decreases → I_B increases

I_B increases → I_C increases

I_C increases → V_C decreases

if V_{CE} goes $\approx 0,2\,V$ → SAT

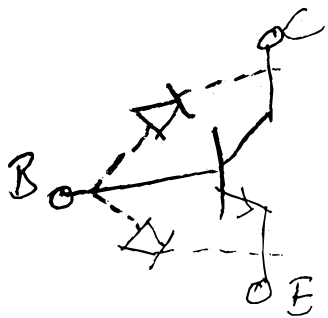
Sat. Conditions

$I_B > 0$

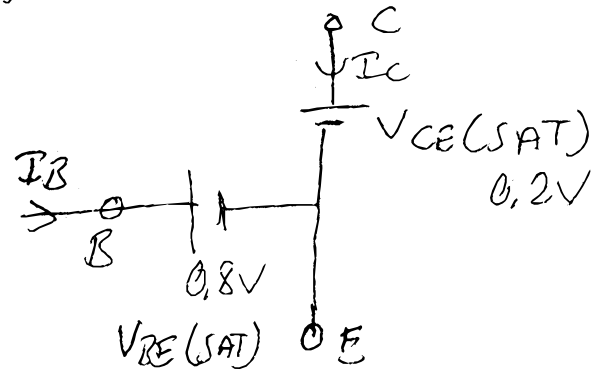
$\beta * I_B > I_C$

$G = \frac{\beta . I_B}{I_C} > 1$

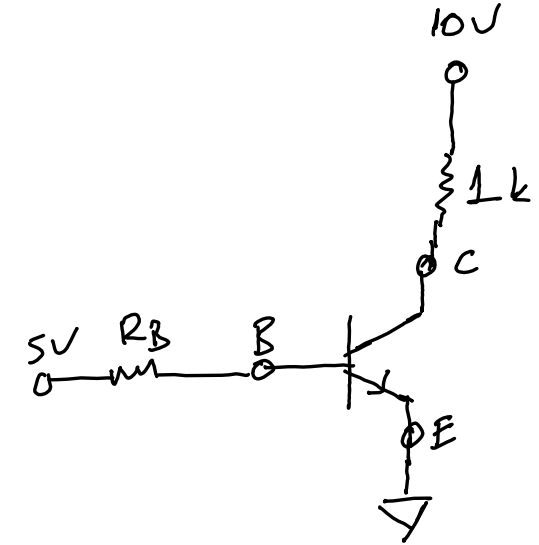
Depth of saturation

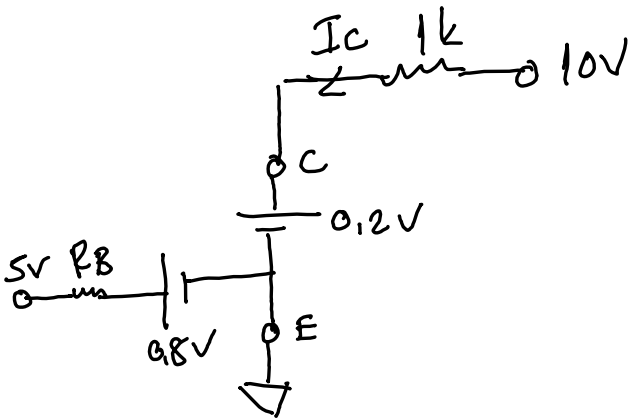






it is fully on state





Find the range for R_B such that Q is in SAT

Let's assume $\beta = 50$.

$V_C = 0,2 \quad I_C = \frac{9,8}{1k} = 9,8\,mA$

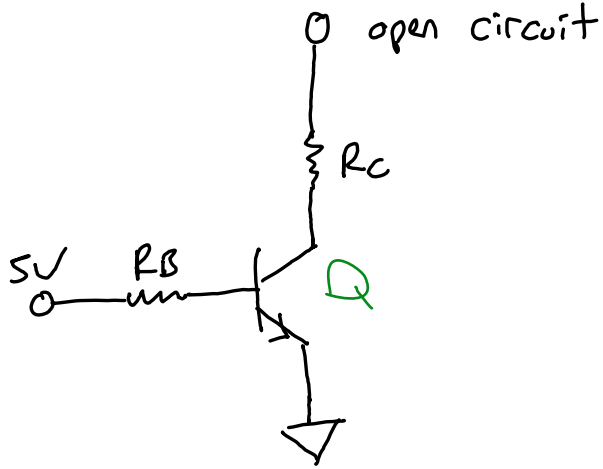
$V_B = 0,8 \quad I_B = \frac{5-0,8}{R_B}$

Check conditions:

$I_B > 0$ ✓ since $R_B > 0$

$\beta \cdot I_B > I_C \rightarrow \frac{50(5-0,8)}{R_B} > 9,8\,mA$

$R_B < \frac{50(5-0,8)}{9,8\,mA} = 21,4\,k\Omega$



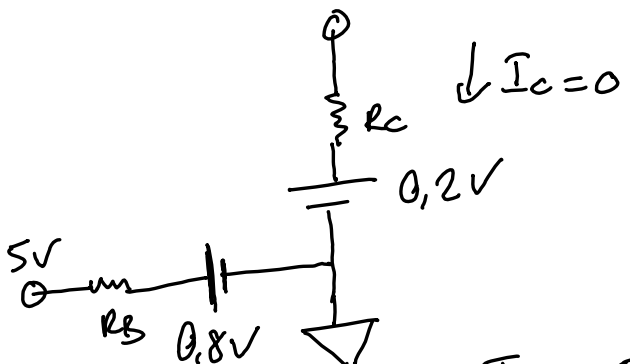
Find R_B such that Q is in SAT

check conditions

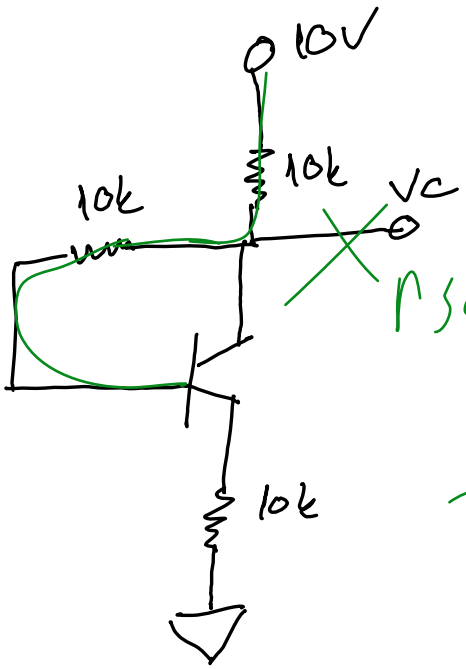
$I_B > 0$ as $R_B > 0$

$\beta I_B > I_c \rightarrow \beta I_B > 0 \rightarrow I_B > 0$

Any positive R_B makes Q into SAT



$$I_B = \frac{5 - 0.8}{R_B}$$



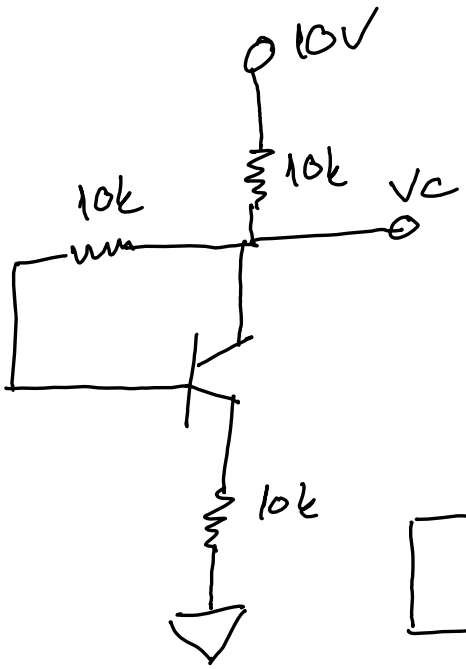
$\beta = 100$ $V_C = ?$

Assume OFF

∂V_C

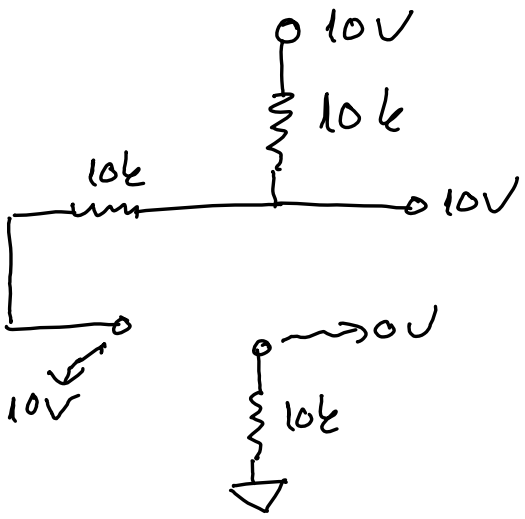
10

90



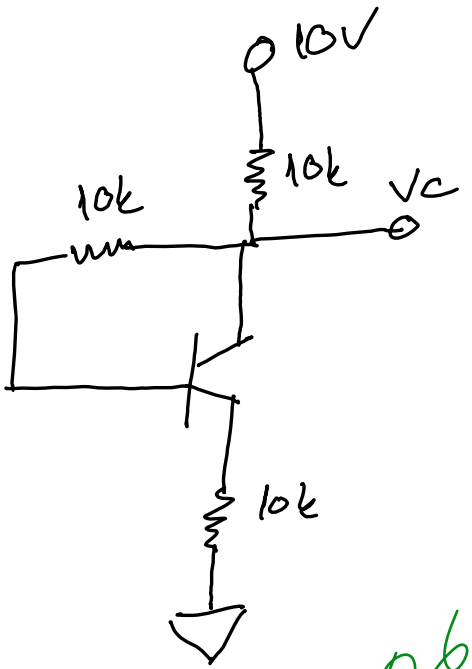
$\beta = 100$ $V_C = ?$

Is it OFF?



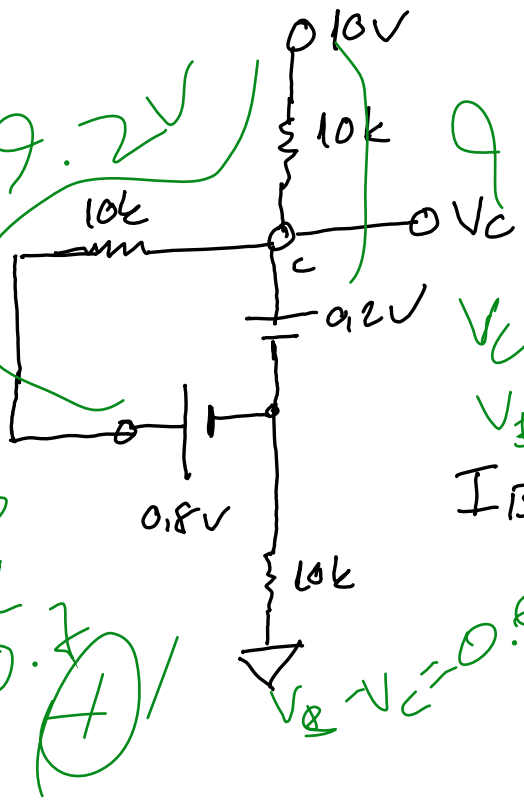
$V_{BE} = 10V$ X

$V_{BE} < 0.7$ for OFF



$\beta = 100$ $V_C = ?$

Is it SAT?



9.2V

9.8V

$V_C = V_E + 0.2V$

$V_B = V_E + 0.8V$

$I_B = \frac{-0.6}{10k} < 0$

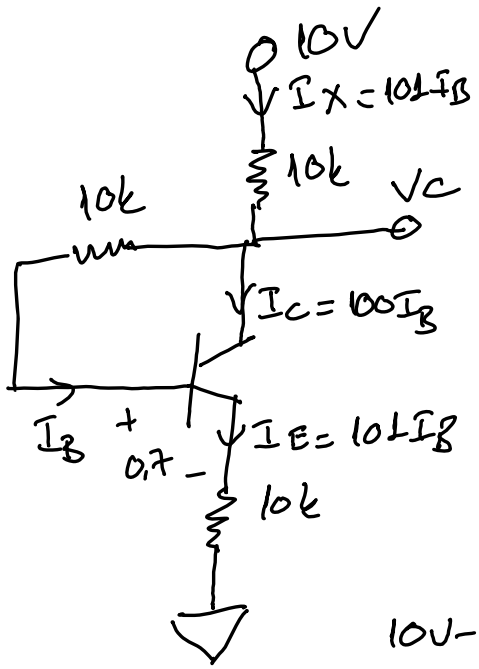
-0.6

$V_C - V_B$

$I_B = \frac{}{10k}$

I_B (0.7) +

$V_B - V_C = 0.6V$ X



$\beta = 100 \quad V_C = ?$

is it FA ?

$$I_X = I_C + I_B = 100 I_B + I_B = 101 I_B$$

$$10V - 101 I_B \cdot 10k - I_B \cdot 10k - 0,7 - 101 I_B \cdot 10k = 0$$

$$203 I_B = 9,3V \Rightarrow I_B = 0,0046\ mA > 0 \checkmark$$

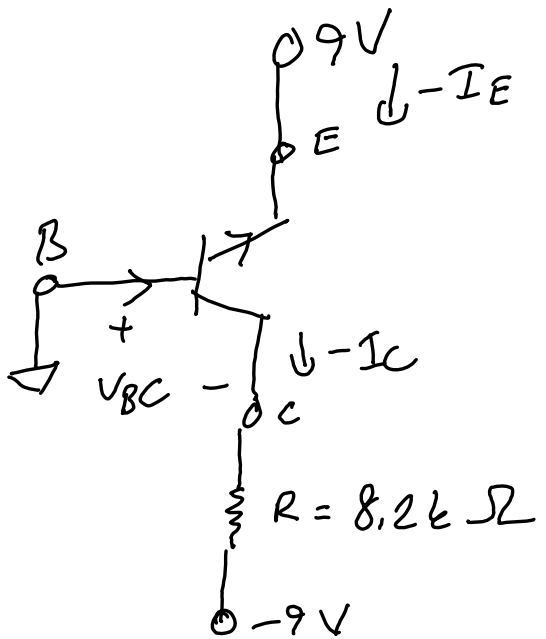
$$V_{BC} = V_B - V_C = -I_B \cdot 10k < 0,7V \checkmark$$

$$V_C = 10 - 101 I_B * 10k = 5,354V$$

Reverse Active Mode

Base - collector diode is forward biased
Base - emitter diode is reverse biased

$$I_E = -\beta_R \cdot I_B \rightarrow I_C = -(\beta_R + 1) I_B$$



$\beta_R = 1$

$V_{BC} = 0,7V$
$V_{BE} = -9V$
} Reverse active

$V_C = -0,7V$

$$-I_C = \frac{-0,7+9}{8.2\,k\Omega} = 1.01\,mA$$

$$I_B = \frac{-I_C}{(\beta_R + 1)} = \frac{1.01}{2} = 0,505\,mA$$

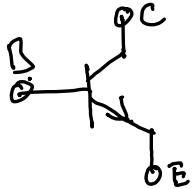
$$I_E = -\beta_R \cdot I_B = -0,505\,mA$$

BJT Applications – Overall!

Forward Active (FA) → Amplifier: D_{BE} on, D_{BC} off.

$I_C = \beta . I_B$, so a small input signal at the base produces a proportionally larger current at the collector. The liner relationship is what amplification is!

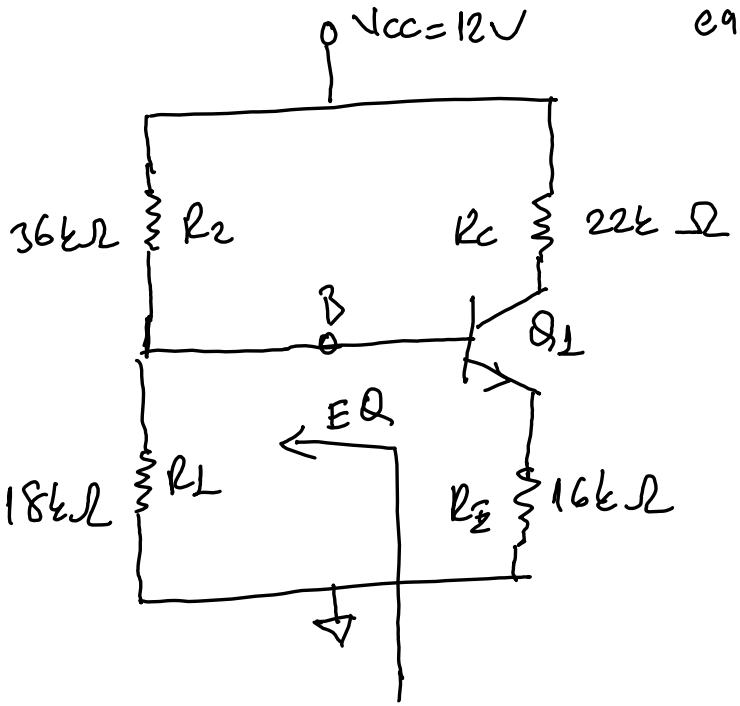
Saturation + Cutoff → Switch: You toggle between the two extremes:



- cutoff: Both junctions off. $I_C \approx 0$, $V_{CE} \approx V_{CC}$. open switch
- saturation: Both junctions on. $I_C \approx \frac{(V_{CC} - 0,2)}{R_C}$, $V_{CE} \approx 0,2V$ closed switch

Reverse active: is useless in practice due to the asymmetric doping of BJT. The current gain is much lower than in FA.

Four Resistor-Bias Network



We would like to find Thevenin eq. circuit seen between B and Ground

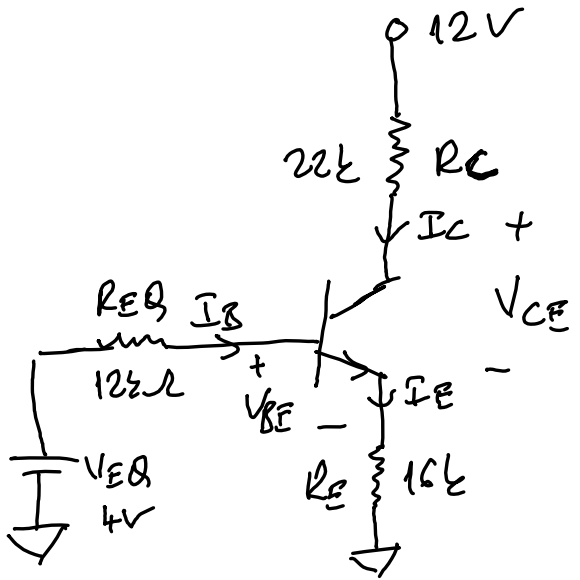
$$V_B = V_{EQ} = \frac{R_1}{R_1 + R_2} \cdot V_{cc}$$

$$= \frac{18}{18+56} \cdot 12 = 4V$$

$$R_B = R_{eq} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$= \frac{18.56}{18+56}$$

$$= 12\Omega$$



At this point, we need to assume a region of operation to be able to continue with the analysis

In this circuit, V_{BE} and V_{CB} are positive, so the assumption of FAR a good one.

$$4 - 12k \cdot I_B - \underbrace{V_{BE}}_{0.7} - 16k \underbrace{I_E}_{(\beta+1)I_B} = 0$$

$\beta = 75$

$$I_B = \frac{4V - 0.7V}{1.21 \cdot 10^6 \Omega} = 2.73\,\mu A$$

$$I_C = \beta I_B = 205\,\mu A$$

$$I_E = (\beta+1) I_B = 208\,\mu A$$

Conditions

$I_B > 0$ ✓

$V_{BC} = V_{BE} - V_{CE}$

$= 0.7 - 4.17$

$= -3.47$ ✓

$$V_{CE} = V_{CC} - R_C \cdot I_C - R_E \cdot I_E = V_{CC} - \left(R_C + \frac{R_E}{\alpha}\right) I_C$$

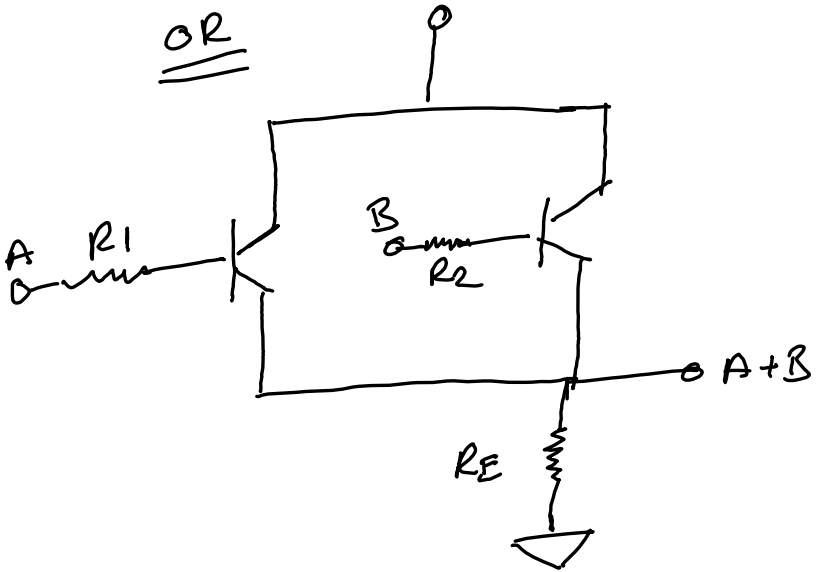
$= 12V - 38.2k \cdot I_C = 12V - 7.83V$

$= 4.17V$ ✓

$\rightsquigarrow I_E \cdot \alpha = I_C$

Logic Gates

OR



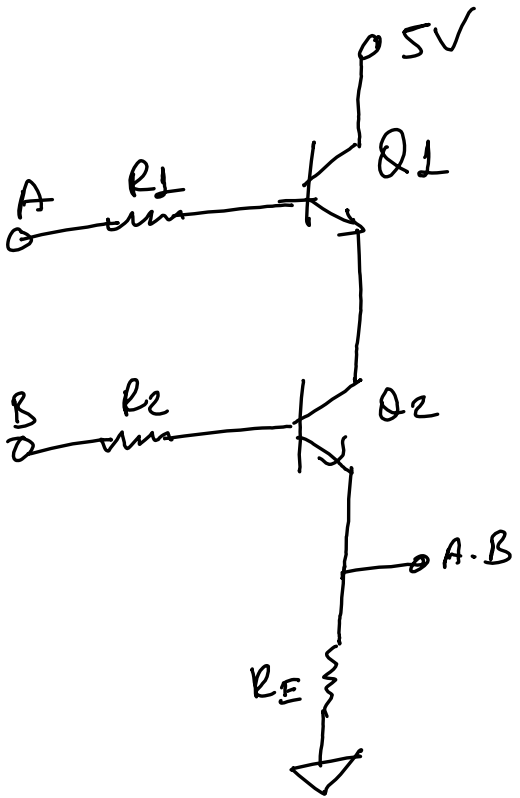
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

Definition:

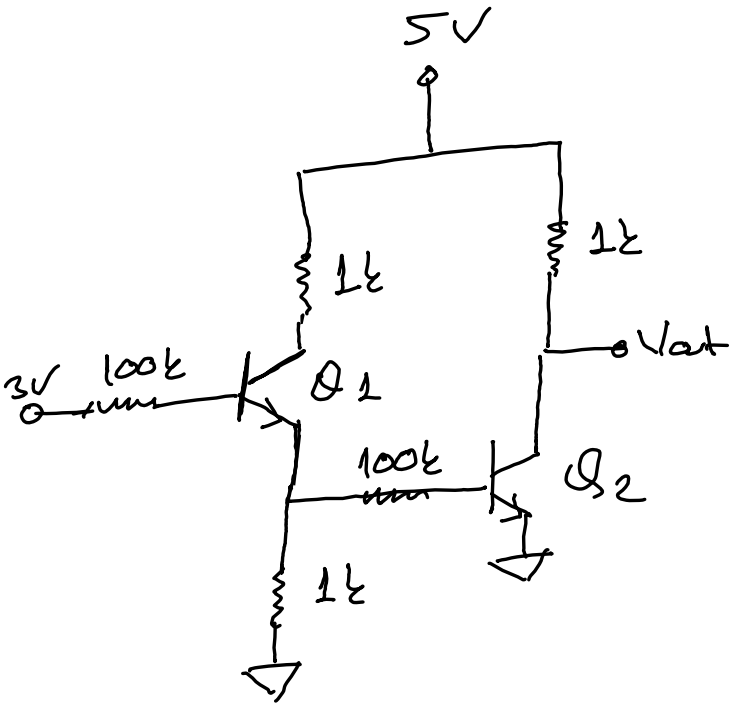
A: 1 → high voltage at the base

A: 0 → 0V at the base

AND



A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

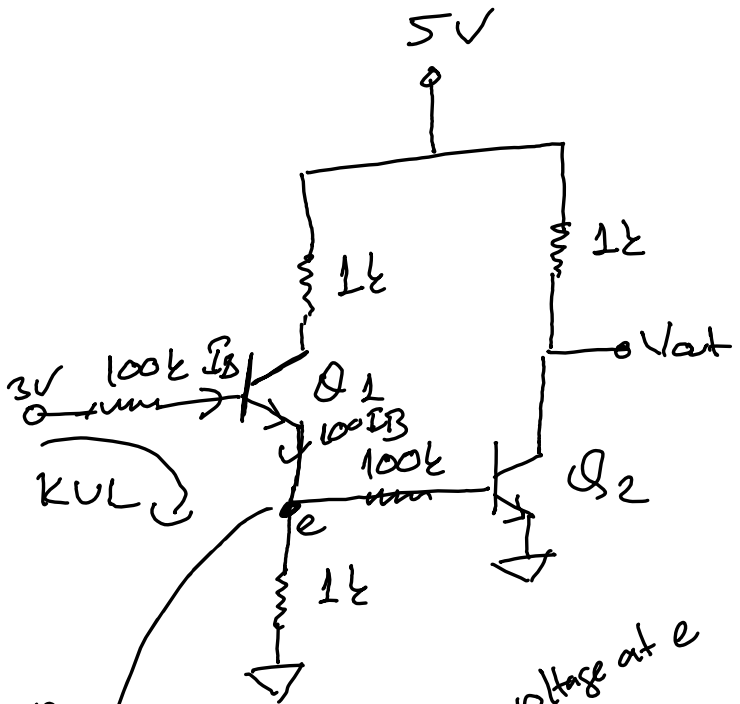


$\beta = 99$

$V_{BE(ON)} = V_{BC(ON)} = 0.7$

$V_{CE(SAT)} = 0.2\,V$

$V_{BE(SAT)} = 0.8\,V$



$e = 3V - 100 I_{B_1} - 0,7$

$= 1.148V$

$V_{C_1} = 3,85$

$V_{B_1} = 1,85$ $\Big\}$ $V_{BC_1} = -2V$ ✓

$I_{B_1} > 0$ ✓

FA for Q1 ✓

given that $\beta = 99$

KCL (voltage at e)

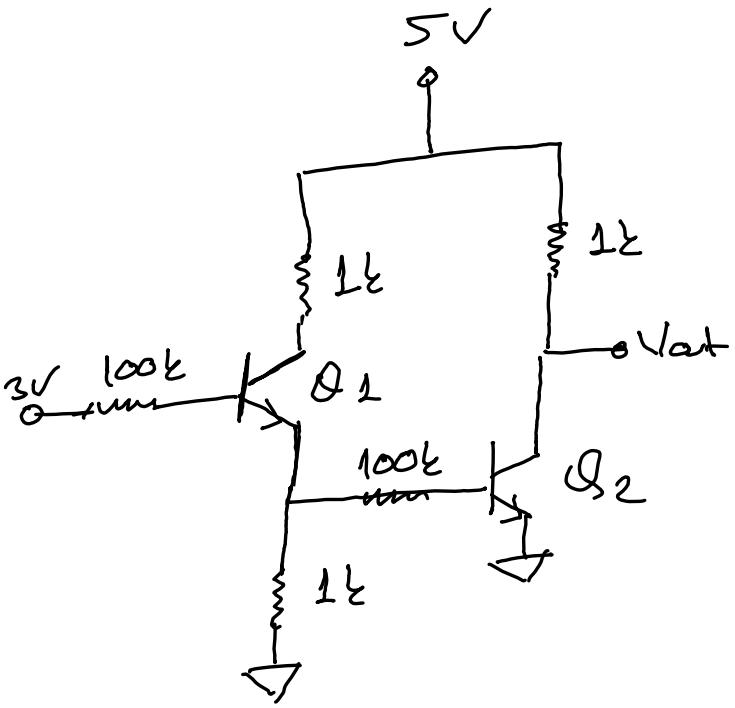
$$\frac{100 I_{B_1} - e}{1K} - \frac{(e - 0,7)}{100K} = 0 \Rightarrow 10^4 \cdot I_{B_1} \cdot K + 0,7 = 101e$$

KVL

$$3 - 100K \cdot I_{B_1} - 0,7 - e = 0$$

$$3 - 100K \cdot I_{B_1} - 0,7 = \frac{10^4}{101} \cdot K I_{B_1} + \frac{0,7}{101}$$

$I_{B_1} = 0,0215 mA$ ✓



$\beta = 99$

$V_{BE(ON)} = V_{BC(ON)} = 0,7$

$V_{CE(SAT)} = 0,2\,V$

$V_{BE(SAT)} = 0,8\,V$

$e = 1.148$

$I_{B_2} = \frac{1.148 - 0,7}{100K} = 0,0045mA$ ✓

$V_{C_2} = 5 - 1k \cdot 99 I_{B_2}$

$\cong 4.553\,V$

$V_{B_2} = 0,7$

$V_{BC_2} = 0,7 - 4.553 < 0,7$ ✓

Q_2 FA

✓